

Market structure and prices in online betting markets: theory and evidence

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Abstract

It is well known that odds offered by bookmakers display a favourite-longshot bias, such that bets on longshots lose more than bets. Explanations commonly focus on bettors having risk-loving preferences or failing to judge small probabilities correctly. We show that favourite-longshot bias can emerge from bookmaking markets being imperfectly competitive because when people disagree about the probabilities of the different possible outcomes of an event, demand for bets on longshots is less sensitive to the odds than bets on favourites. We compare odds on over 150,000 European soccer games from two different bookmaking markets that differ in their level of competition and report predictions consistent with our framework. In particular, the data confirm a distinctive prediction from our imperfect competition model about returns from betting on draws.

Keywords: market structure; favourite-longshot bias; sports betting

JEL classifications: D41, D42, G14, L83

1. Introduction

Online sports betting has become a huge global business. Mobile internet technologies have hugely lowered the transaction costs for placing bets and legal developments, such as the 2018 US Supreme Court decision voiding previous federal restrictions on sports betting, have opened up new markets. This article presents a theoretical framework for understanding the modern online betting business and provides evidence to support this framework.

Theoretical research on betting on sports has largely concentrated on pari-mutuel betting, which pools all bets and pays them out (minus a fraction to cover costs and profits) to those who picked the winner in proportion to the size of their bet. This focus reflected that pari-mutuel betting at horse racing tracks was exempted from US prohibitions on sports betting and was thus the best-known way to bet in the US. However, the modern online sports betting business features fixed odds set by bookmakers. They make offers such as ‘You take back \$3 if your pick wins and lose your \$1 bet otherwise’, and if a bettor accepts this offer, their odds are not changed by subsequent actions of other bettors.

This literature has often examined the favourite-longshot bias in pari-mutuel markets, the empirical pattern that bets on longshots tend to lose more than bets on favourites. Because pari-mutuel payouts are determined by betting volumes, theoretical explanations of this bias have focused on the behaviour of bettors. Explanations included that gamblers

may have locally risk-loving preferences and enjoy the high-variance returns from betting on low-probability outcomes (Quandt 1986), that they systematically overestimate probabilities of unlikely events (Thaler and Ziemba 1988) or that there may be ‘noise gamblers’ who pick their bets randomly (Hurley and McDonough 1995).¹

These explanations could also explain why some studies find favourite-longshot bias patterns in fixed-odds betting markets if their odds were set similarly as pari-mutuel markets. Going back to Pankoff (1968), many studies have assumed that fixed-odds bookmakers adjust odds so they obtain a risk-free return due to payouts being the same no matter which contestant wins, meaning odds end-up being the same as they would have been under a pari-mutuel structure. However, there is substantial evidence that this is not how bookmakers set odds. Jay Kornegay is the most famous bookmaker in Las Vegas. Mieh (2019) quotes him: *‘I think there’s a little urban legend that’s out there, that we just want to balance every game ... You go through every single game today and probably none of them ... are balanced on both sides. That scenario right there, it just doesn’t happen ... What we’re trying to do is put ourselves in the best possible position to win, not necessarily balancing both sides.’* Studies documenting that bookmakers do not ‘balance their books’ include Strumpf (2003), Levitt (2004), and Flepp, Nüesch and Franck (2016).

In this article, we show how the pattern of odds observed in fixed-odds betting markets can be explained without appealing to unorthodox assumptions about the behaviour of bookmaking firms or ad hoc ideas about the behaviour of bettors, such as them being risk lovers or systematically misjudging small probabilities. Rather than making a special assumption for bookmakers, our approach assumes they are businesses that make a normal rate of return if the market is competitive and maximize their profits based on the elasticity of demand if they are monopolists. We assume betting markets exist because people disagree about the probabilities of the outcomes of sporting events and that there is a form of ‘wisdom of crowds’ so that the beliefs of potential bettors are centred around the true probabilities. Ali (1977) showed this assumption implied favourite-longshot bias in pari-mutuel betting markets.² We show that disagreement among bettors will not generate favourite-longshot bias in a competitive fixed-odds market but will produce this bias when the markets have an imperfectly competitive structure.

This article is structured as follows. First, we consider a perfectly competitive fixed-odds bookmaking market with a normal profit requirement in which odds do not exhibit favourite-longshot bias.

Second, we show that monopolistic bookmakers will set odds that feature favourite-longshot bias. As shown by Montone (2021), monopolist bookmakers set odds as a ‘markdown’ on their zero-profit perfectly competitive levels with the size of the markdown depending on the elasticity of demand of bettors: Bettors that are more price-sensitive get quoted better odds. We show that disagreement among bettors implies the demand for bets on favourites is more sensitive to odds than demand for bets on longshots and our model predicts that expected loss rates on bets rise nonlinearly as the probability of winning declines.

The result about price sensitivity of demand for bets can be illustrated as follows. Consider an event with two competitors where the probability of the favourite winning is $p = 0.7$ and bettors have beliefs $\tilde{p}$ about p . Let O_F and O_L be the so-called decimal odds for the favourite and longshot (the payout on a successful \$1 bet inclusive of the original bet). The ‘fair odds’ at which bets break even on average are $O_F = \frac{1}{0.7} = 1.43$ and $O_L = \frac{1}{0.3} = 3.33$. Assume bettors make bets if they believe the expected payout is at least one and suppose the bettors that are most optimistic about the favourite have $\tilde{p} = 0.8$ while

¹ Ottaviani and Sørensen (2008) is a comprehensive review of the various theories on favourite-longshot bias.

² Gandhi and Serrano-Padial (2015) generalized these results to illustrate how disagreement generates favourite-longshot bias in pari-mutuel betting as long as the true probability is part of set of beliefs.

those who are most optimistic about the longshot have $\tilde{p} = 0.6$. Those who are most optimistic about the favourite's prospects would accept odds of $O_F = \frac{1}{0.8} = 1.25$, which is one-eighth lower than competitive odds. Conversely, those who are most optimistic about the longshot would accept odds of $O_L = \frac{1}{0.4} = 2.5$ which is one-quarter lower than competitive odds.

Third, we present a version of our monopoly model designed to match betting odds on soccer games where there are three possible outcomes: Home win, away win or draw. The pattern of favourite-longshot bias that we derive for sports with two possible outcomes fits well with the evidence for fixed-odds betting on two-outcome sports such as tennis but it is possible that some other factor than disagreement among bettors generates this finding. For example, longshot bettors may be less elastic because of their preferences for risk rather than the disagreement-based mechanism that we have outlined.

An advantage of examining betting on soccer outcomes is that our model generates concrete predictions for loss rates on draw bets that are unlikely to be explained by other theories. Specifically, the model predicts that when draws are most likely to occur, disagreement among bettors will be low and loss rates on these bets will be as low as for extreme favourites. It also predicts that, when draws are relatively more likely, average loss rates from betting on draws should be lower than for betting on longshots to win even if the bets have equal probabilities of winning. Conversely, when draws are relatively less likely, average loss rates from betting on draws should be higher than for betting on longshots to win even if the bets have equal probabilities of winning.

Fourth, we use these results to explain betting odds for over 150,000 European soccer games. We show that the market for betting on home/away/draw outcomes exhibits substantial favourite-longshot bias, as found in previous studies with smaller samples such as Buhagiar, Cortis, and Newall (2018) and Angelini and De Angelis (2019). These losses closely match the nonlinear pattern predicted by our model. In addition, the data confirm our monopolistic model's very specific predictions about loss rates from bets on draws.

We also compare our results for home/away/draw bets with results from bets placed on the same set of games in the so-called Asian Handicap market, where payouts on bets depend on an adjustment of the match result that applies a deduction to the goals total of the team considered more likely to win. The Asian handicap market is known to be popular with professional gamblers and betting syndicates who place large-volume bets with 'sharp' bookmakers such as Pinnacle who compete against each other to offer the most attractive odds. These bookmakers make low profit margins per bet but offset this by generating high volume. In contrast, the market for home/away/draw bets is dominated by retail European bookmakers who have high margins and engage in various anti-competitive practices such as limiting or banning customers who appear to be actively seeking out the best odds. These firms rarely compete on odds offered, focusing instead on marketing and promotions to attract and retain high-revenue customers.

Our models suggest the market dominated by the bookmakers who operate in an essentially competitive manner should not exhibit favourite-longshot bias while the market dominated by firms with anti-competitive practices will feature this bias. The evidence confirms what the models suggest. The Asian handicap market does not exhibit favourite-longshot bias for the same set of games for which a significant bias is reported for home/away/draw betting. By its nature, the handicap market features fewer extreme longshots or favourites but within the probability ranges where you can compare both types of bets, the pattern for loss rates is very different.

2. Competitive fixed odds bookmakers

We first consider a competitive fixed odds bookmaking market in which there is an event with N possible outcomes, each with a probability p_i of happening. Bookmakers know the

values of p_i and offer decimal odds O_i so that a \$1 bet returns O_i back to the gambler if event i happens. There are a number of risk-neutral bookmakers who compete against each other for business and the competitive equilibrium requires each to make an expected profit rate of θ on each bet, reflecting the need to earn a ‘normal’ rate of profit. If expected profit rates are higher than this, new bookmakers enter the market and offer lower odds to bring the profit rate back to θ .³ Costs for bookmakers are a fraction μ of the total dollar amount of bets placed.⁴

The competitive equilibrium profit rate of θ implies the following equality for revenue minus costs on a \$1 bet

$$1 - \mu - p_i O_i = \theta \quad (1)$$

so the equilibrium odds are

$$O_i = \frac{1 - \mu - \theta}{p_i} \quad (2)$$

This means that the ratio of the odds for two bets reflects the ratio of the underlying probabilities. Bettors can also figure out the value of $\mu + \theta$ from these odds because

$$\sum_{i=1}^N \frac{1}{O_i} = \sum_{i=1}^N \frac{p_i}{1 - \mu - \theta} = \frac{1}{1 - \mu - \theta} \quad (3)$$

This means

$$\mu + \theta = 1 - \frac{1}{\sum_{i=1}^N \frac{1}{O_i}} \quad (4)$$

In other words, gamblers can use the ‘overround’ from the quoted odds—the sum of the inverse of the odds—to calculate the bookmaker’s gross profit margin and with this value calculated, bettors can also figure out the underlying probabilities from the odds using Equation (1). As discussed in [Hegarty and Whelan \(2025\)](#), most practical advice on sports betting recommends bettors do this calculation to assess the gross profit rate of bookmakers on bets on an event. Also, once they have calculated $\mu + \theta$, bettors can back out the values of the underlying probabilities from the odds.

Without describing how potential gamblers behave, we can see that the theoretical devices that have been used to explain favourite-longshot bias in pari-mutuel markets will not influence odds in this market. Gamblers may have locally risk-loving preferences and enjoy making high-variance low-probability bets, as suggested by [Quandt \(1986\)](#). Alternatively, gamblers may systematically overestimate the probabilities of unlikely events, as suggested by [Kahneman and Tversky’s \(1979\)](#) prospect theory and discussed in the context of soccer betting by [Franke \(2020\)](#). There could be ‘noise bettors’ who pick their bets randomly, as modelled by [Hurley and McDonough \(1995\)](#) or perhaps gamblers who choose to back a team because they are a fan.

³ [Whelan \(2024\)](#) presents a model of competitive bookmakers with risk averse preferences so that risk is taken into account when deciding whether to enter or leave the market. We also discuss extending our monopolistic model to incorporate risk aversion for bookmakers later in this paper.

⁴ This per bet cost parameter μ is likely, of course, to vary across events. For example, the costs involved in setting odds for a match which attracts a lot of betting are likely to be similar to the cost for a match which receives few bets. As such, per-bet costs for bookmakers are likely to be lower for popular events.

Each of these features drives betting volume towards longshot bets in a way that is unrelated to the fundamental chance the bet has of winning. Because the pari-mutuel structure requires winning odds to be inversely related to betting volumes, this reduces the winning pari-mutuel odds on the longshot team and increases them on the favourite. However, these factors would have no influence on odds in a competitive market because these odds are not determined by betting volumes. In each case, these mechanisms imply the existence of gamblers who would back the longshot at lower odds than the competitive prices. But getting those gamblers to accept these poorer odds would generate rates of return for bookmakers above θ and incentivize other bookmakers to undercut those making these excess profits.

The form of disagreement among bettors that we discuss below—which follows Ali (1977) in assuming that bettors have symmetric beliefs about probabilities that centre on their true values—will also not induce favourite-longshot bias in competitive markets. Ali explained how pari-mutuel betting odds for favourites needed to be more attractive in terms of expected returns for the average bettor than for longshots to attract the additional volumes required for odds on favourites to be lower than for longshots. In contrast, in competitive markets, odds do not depend on betting volumes, so this kind of disagreement would have no impact.

3. Disagreement with monopolistic bookmakers

We now consider the case where the bookmaker is a monopolist. Our model shares similarities with Shin (1991) in that it has a monopolistic bookmaker and disagreement among bettors generates demand functions for bets on each outcome. There are, however, two important differences. First, we do not incorporate ‘insiders’ who know the outcome of the sporting event before it has happened. The motivation for this feature stemmed from the early literature on sports betting which was focused largely on horse racing, where sometimes those who are involved in training a horse will know it has a much higher or lower chance of winning than the general public. In the high-profile sports that dominate modern online betting, this level of inside information is generally unlikely to apply.

Second, Shin’s model featured non-insider bettors with no useful knowledge about the sporting event. For a two-team event, he assumed these bettors had beliefs about the probability that one of the teams will win that were uniform on $[0, 1]$ no matter what the true value of this probability is. In contrast, we will assume the betting public is, on average, correct in its assessment of the underlying probabilities.

Of course, in the real world, there are usually several competing bookmakers. In the Supplementary Appendix, we show that the odds that emerge from a Cournot oligopoly model with demand due to disagreement among bettors will lie between the competition and monopoly model outcomes, depending on the number of firms. We present the monopoly model because it makes clearer how imperfect competition produces a favourite-longshot bias, with the key mechanism being differences in elasticity of demand for different bets.

3.1 How the monopolist bookmaker sets odds

Consider again a contest with N possible outcomes, each with a probability p_i of happening. A monopolistic bookmaker, who knows the probabilities, has a processing cost for each bet equal to a fraction μ of the amount placed and faces a demand function for bets $D(O_i)$ such that $D'(O_i) > 0$.

At first sight, bookmaking seems a very different business to the typical monopolist in economics. The prices set (the betting odds) have only an indirect influence on its revenues (since higher odds attract more bets) but have a direct influence on its costs. However, Montone (2021) has shown that profit-maximizing odds for monopoly bookmakers follow

the same logic as the traditional economic theory of monopoly pricing. This result is derived as follows. The bookmaker's expected profit on bets placed on outcome i is

$$E(\Pi_i) = (1 - \mu - p_i O_i) D(O_i) \quad (5)$$

The condition for maximizing expected profit is

$$\frac{dE(\Pi)}{dO_i} = (1 - \mu - p_i O_i) D'(O_i) - p_i D(O_i) = 0 \quad (6)$$

This solves to give

$$O_i = \frac{1 - \mu}{p_i} - \frac{D(O_i)}{D'(O_i)} \quad (7)$$

We can see immediately that the odds offered by a monopolist will be lower than in a zero-profit competitive market. Defining the elasticity of demand for the bet as

$$\varepsilon_i = \frac{O_i D'(O_i)}{D(O_i)} \quad (8)$$

optimal odds become

$$O_i = \frac{\varepsilon_i}{\varepsilon_i + 1} \frac{1 - \mu}{p_i} \quad (9)$$

The bookmaker chooses to set odds as a 'markdown' on the zero-profit competitive odds and bets with higher elasticities of demand will have lower markdowns, with odds tending towards their zero-profit-margin competitive level as the elasticity of demand becomes infinite. If we interpret the price set by the monopolist as the inverse of the odds, we recover the standard result that the monopoly price is a markup over the competitive price with the usual markup formula.

3.2 Disagreement and elasticity of demand

So what determines elasticity of demand and thus betting odds? We use a simple model of bettors who disagree about the probabilities of outcomes and show that the demand for longshot bets is less sensitive to prices than demand for bets on favourites. We assume there is a continuum of risk-neutral potential gamblers of size 1 who can choose to bet or not. Any bets that are placed are normalized to be of size one. These are standard assumptions used in almost all theoretical work on betting markets. In Section 7, we will discuss how these assumptions could be generalized to allow for risk-averse bettors and bet sizes framed as a fraction of wealth.

Potential gamblers have a subjective belief $\tilde{p}_i$ about the true probability p_i that a sporting event ends with the i -th outcome. As in Ali (1977), beliefs are characterized by a cumulative distribution function $F(\tilde{p}_i)$ with mean p_i , meaning the gamblers are on average correct about the probability of the event occurring, and beliefs are symmetrically distributed around p . Potential bettors are risk-neutral and only bet if the odds imply an expected profit greater than or equal to zero, so they take the bet on outcome i if

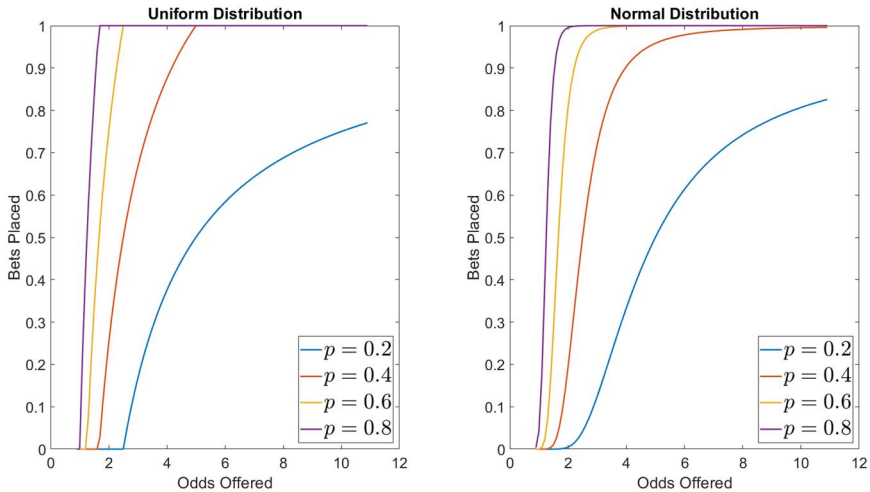


Figure 1. Total bets placed as a function of odds and underlying probability of bet win p . The uniform belief distributions are uniform on $[p - 0.2, p + 0.2]$. The normally distributed beliefs are of the form $N(p, 0.115)$. Source: Authors' calculations.

$$\tilde{p}_i O_i \geq 1 \quad (10)$$

In the introduction, we described the intuition for why demand for longshot bets is less sensitive to changes in odds than demand for favourite bets. A concrete example of a distribution of beliefs can give a fuller illustration of this result. Consider again an event with two competitors where the probability of the favourite winning is $p = 0.7$. Assume now potential bettors' beliefs $\tilde{p}$ about p are uniformly distributed on $[0.6, 0.8]$ so that, on average, the public's belief about this probability is correct. As noted above, the 'fair odds' in a market with no gross profits for bookmakers would be $O_F = \frac{1}{0.7} = 1.43$ and $O_L = \frac{1}{0.3} = 3.33$. At these odds, half the potential bettors (those with $\tilde{p} \geq 0.7$) will take the bet on the favourite and the other half will take the bet on the longshot.

Now suppose the odds on both teams were cut by 10 per cent. At the new favourite odds of $O_F = 1.29$ only those with $\tilde{p} > 0.777 = \frac{1}{1.29}$ will take the bet, meaning only 11 per cent of potential bettors now bet on the favourite. In contrast, at the new longshot odds of $O_L = 3$, those with $1 - \tilde{p} > \frac{1}{3}$, meaning $\tilde{p} < 0.667$ will take the bet, meaning 33 per cent of potential bettors will still take the longshot bet. This illustrates the weaker price sensitivity of demand for longshot bets.

The result about demand for low p bets being less elastic does not rely on the uniform distribution assumption. Figure 1 shows demand for bets as a function of odds on an outcome with win probability p_i for different values of p_i and using both uniform distributions for beliefs on $[p_i - 0.2, p_i + 0.2]$ and normal distributions with mean p_i and standard deviation 0.115 (the same standard deviation as the uniform distribution). For both normal and uniform distributions, we see demand functions for bets on high probability events are much steeper than for bets on low probability events.

3.3 Odds with a uniform distribution

To provide a concrete illustration of how the monopolist sets odds, we will stick with beliefs about the probability p_i being uniformly distributed over $[p_i - \sigma_i, p_i + \sigma_i]$. This implies a demand function for bets on outcome i of the form

$$D(O_i) = 1 - \frac{1}{2\sigma_i} \left[\frac{1}{O_i} - p_i + \sigma_i \right] \quad (11)$$

And expected profit is

$$E(\Pi_i) = (1 - \mu - p_i O_i) \left[1 - \frac{1}{2\sigma_i} \left[\frac{1}{O_i} - p_i + \sigma_i \right] \right] \quad (12)$$

Differentiating with respect to O_i and solving, the optimal odds are

$$O_i = \sqrt{\frac{1 - \mu}{p_i(p_i + \sigma_i)}} \quad (13)$$

Odds depend negatively on σ_i because rising disagreement reduces the elasticity of demand by increasing the fraction of bettors willing to take bets at relatively bad odds.

One can easily show that these odds generate a favourite-longshot bias pattern of returns on longshot bets being worse than for favourite bets because the expected payout on bets is

$$p_i O_i = \sqrt{\frac{(1 - \mu)p_i}{(p_i + \sigma_i)}} \quad (14)$$

which depends positively on p_i and negatively on σ_i . To illustrate the magnitudes of this favourite-longshot bias, suppose $\mu = 0.02$ and $\sigma_i = 0.06$. The expected loss rate ($p_i O_i - 1$) for a bettor that chooses a bet with probability of success $p_i = 0.8$ is 4.5 per cent. The corresponding expected loss for $p_i = 0.2$ is 13.2 per cent.

Inserting the optimal odds into the demand function, the total amount of bets is

$$D(O_i) = 1 - \frac{1}{2\sigma_i} \left[\sqrt{\frac{p_i(p_i + \sigma_i)}{1 - \mu}} - p_i + \sigma_i \right] \quad (15)$$

This declines as p_i rises because the effect on the square root term is smaller than the effect on the linear term being subtracted. So, despite being offered comparatively worse odds with greater losses per bet, the quantity of bets on longshots is greater than on favourites. Betting volumes also rise with σ_i because the more disagreement there is, the more people there are who believe they are going to beat the bookmaker's margin and make a profit.

The upper panels of Fig. 2 illustrate how the monopolist's odds differ from zero-profit competitive odds (so $\theta = 0$) for various values of p_i ranging from 0.1 to 0.9 and for uniform beliefs on $[p_i - \sigma_i, p_i + \sigma_i]$ with $\sigma_i = 0.06$ and $\mu = 0.02$ (we will explain these parameter choices later). The monopolist's odds decline substantially relative to competitive odds as the true probability of the bet winning falls. The bottom left panel shows how the equilibrium elasticity of demand rises sharply as p_i goes up, meaning smaller markdowns on competitive odds for bets that are more likely to win. The bottom right panel shows the expected loss rates for bets as a function of the bet's probability of winning. There is a highly nonlinear pattern of rising loss rates as the probability of winning falls. For this calibration, longshot bets with win probabilities as low as 0.1 have expected loss rates of over 10 per cent.

3.4 Two outcome events

The derivation just presented assumed the extent of disagreement about probabilities differed for each possible outcome. However, with two possible outcomes, if σ represents the amount of disagreement about the probability of the favourite winning, it also represents the

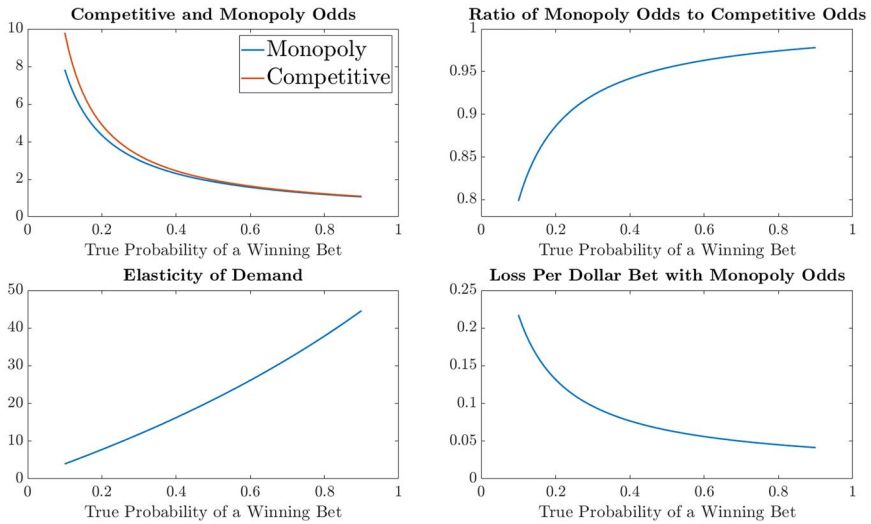


Figure 2. Comparing competitive and monopoly odds ($\sigma = 0.06, \mu = 0.02, \theta = 0$).

Source: Authors' calculations.

amount of disagreement about the probability of the longshot winning. Thus, for a two-team game in which one team is the favourite with a probability of winning $p > 0.5$ and the public's beliefs are uniform on $[p - \sigma, p + \sigma]$, the ratio of favourite to longshot odds will be

$$\frac{O_F}{O_L} = \sqrt{\frac{(1-p)(1-p+\sigma)}{p(p+\sigma)}} \quad (16)$$

Note that if there is no disagreement in beliefs, so $\sigma = 0$, this formula gives the same odds as the competitive market. However, in this model, the bookmaking market would collapse in this case because all bettors would have a correct assessment of the value of p and none of them would believe they could make a profit given the bookmaker's need to cover costs.

The square-root formula in Equation (16) looks similar to a formula in Shin's (1991) paper for the ratio of odds on the favourite to odds on the longshot

$$\frac{O_F}{O_L} = \sqrt{\frac{1-p}{p}} \quad (17)$$

but the underlying assumptions generating the formulas are different. The bettors in our model are correct, on average, about the probability of each team winning whereas the 'outsiders' in Shin's model have uniform beliefs over $[0, 1]$. Only when $p = 0.5$ and $\sigma = 0.5$ do the two formulas imply the same odds.

We believe our assumptions are a more reasonable representation of bettors in modern online sports markets. When considering Liverpool versus Southampton, there may be many people who do not have an accurate opinion on the likely outcome. However, the population of people who are willing to consider making a bet on this game will surely have opinions that are somewhat informed by reality (which, at the time of writing, is that Liverpool are much better than Southampton). Given this, we view the assumption that the set of potential bettors are on average correct in their assessment as an appropriate

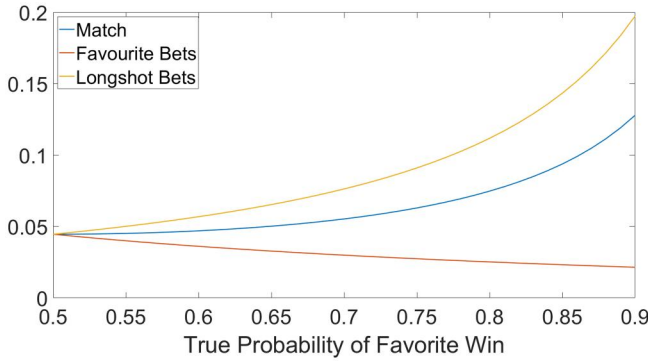


Figure 3. Bookmaker profit rates on bets on two-team event for $\mu = 0.02$, $\sigma = 0.06$.
Source: Authors' calculations.

modelling approach. One can also show that Shin's formula implies a much larger favourite-longshot bias than is observed in datasets such as the one we use for European soccer later in this article as well as a much higher average profit margin for the bookmaker.⁵

The pattern in which payout rates on longshots fall nonlinearly as p rises means bookmakers make a higher profit rate on games that are mis-matches than on games that are closer to toss-ups. Figure 3 illustrates how the nonlinear pattern of increased loss rates for longshots (and thus increased profits for bookmakers) means the additional gains from bets on more extreme longshots offset the slightly lower profits obtained from bets on a favourite with a high win probability.

One question about this result is whether gamblers can tell from the odds that the bookmakers are making larger profits on bets on games with a strong favourite and thus that at least one of the bets may offer poor value. As discussed above, gamblers tend to use the overround implied by the odds to measure a bookmaker's gross profit margin. In a competitive market, the overround is

$$R = \frac{\mu + \theta}{1 - \mu - \theta} \quad (18)$$

which depends positively on bookmakers' costs and the profit margin requirement and would be zero if these didn't exist.

In this model, the overround is

$$R = \frac{1}{O_F} + \frac{1}{O_L} - 1 \quad (19)$$

$$= \sqrt{\frac{p(p + \sigma)}{1 - \mu}} + \sqrt{\frac{(1 - p)(1 - p + \sigma)}{1 - \mu}} - 1 \quad (20)$$

This depends positively on μ and σ . In fact, it turns out the overround increases almost one-for-one with the value of σ and for each value of σ there is very little variation in the overround with the value of p . Indeed, perhaps surprisingly, the model predicts that overrounds for games where there is a strong favourite will be slightly lower than for games

⁵ See Whelan (2025) for a more detailed discussion of the properties of Shin's model.

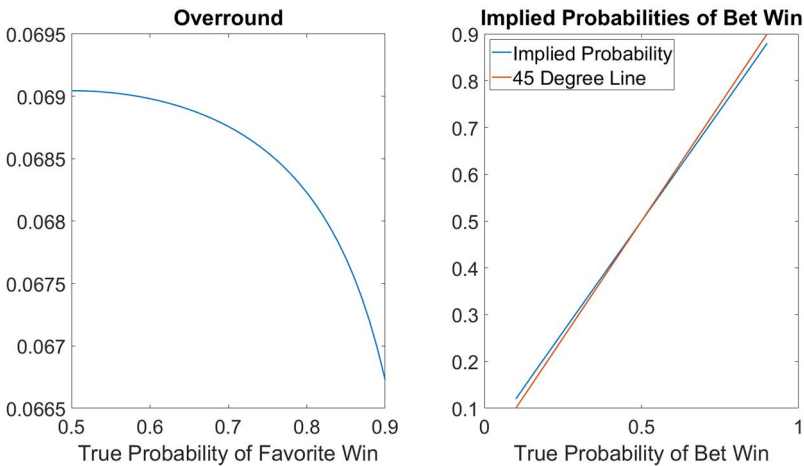


Figure 4. Overround and comparison of true and implied probabilities ($\mu = 0.02, \sigma = 0.06$).
Source: Authors' calculations.

that are effectively toss ups: See the left panel of Fig. 4. This means the overround does not send the correct signal to gamblers about the profit rates being earned by the bookmaker.

This disconnect between the overround and the true profit rate occurs because the implicit model behind using the overround to estimate the bookmaker's gross profit margin—that odds start as the inverse of the probabilities and are then reached marked down proportionately—does not describe how the monopolist bookmaker sets odds. The right panel of Fig. 4 shows the problem with using the overround to estimate gross profit margins and probabilities. The probabilities implied by the standard way of extracting the bookmaker's margin from odds, as described above, do not match with the actual probabilities. Longshots win less often than this method suggests and favourites win more often.

4. A model of betting on soccer

The monopolistic model with two outcomes can be applied to many sporting events, such as tennis matches. Indeed, we show in [Supplementary Appendix A](#) that the predicted loss rate pattern from the model matches observed loss rates from a large dataset for odds and outcomes for professional tennis matches. The model's predicted pattern of lower overrounds as the probability of the favourite winning rises is also confirmed by the tennis data. Our focus, however, will be to use the model to explain the odds on European soccer matches where people can bet on three possible outcomes: A home win, an away win and a draw.

This approach will have two advantages. First, the larger set of outcomes helps to generate some unique predictions of our model that can then be assessed against the data. For two-outcome sports, our disagreement-based model predicts a nonlinear pattern of rising losses as the probability of success falls and explains this as a result of a falling elasticity of demand. However, one could imagine other theories also explaining this pattern: For example, perhaps some bettors innately like risk more than others and this taste for risk makes them less elastic in their demand for longshot bets. The three-outcome model will generate specific predictions due only to the disagreement-based structure of the model. Second, we will be able to compare outcomes from betting in two very different betting markets for the same large dataset of soccer games.

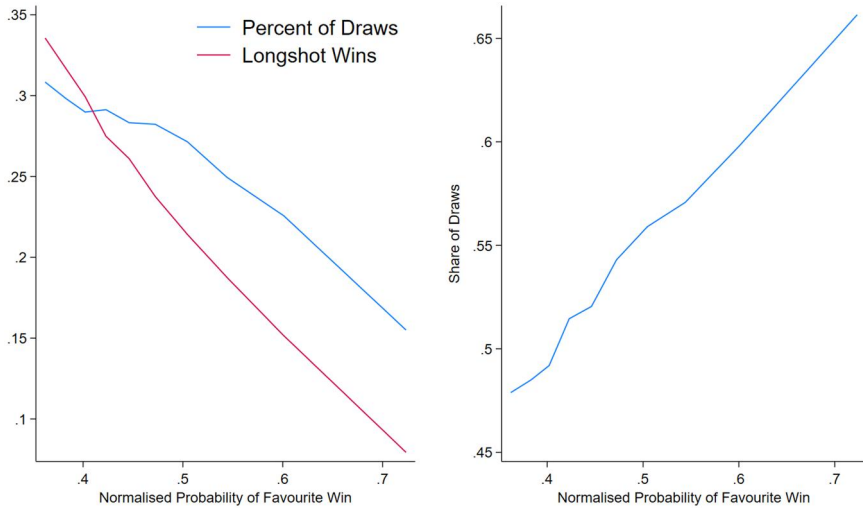


Figure 5. Draws rise relative to longshot wins as the probability of the favourite winning rises.

Source: Authors' calculations based on data from www.football-data.co.uk.

Note: Estimates based on the deciles of our dataset of European soccer games, organized by the ex-ante probability of the favourite winning, as measured by normalized probabilities. Left panel shows fraction of draws and longshot wins for each decile. Right panel shows the share of draws among non-favourite-win outcomes.

4.1 Evidence on the likelihood of draws

Calibrating a model with three outcomes requires specifying beliefs about more than one probability. One way to do this would be to assume a fixed number for the probability of a draw, so the disagreement is only about the probabilities of the home and away wins. Pope and Peel (1989), for example, used data from British soccer matches to argue that while the overall percentage of matches ending in draws was predictable, there was no reliable way to forecast which kinds of matches might end in a draw. That result, however, does not hold in our comprehensive dataset of European soccer matches. In fact, contingent on the probability of one team winning, there is a clear pattern for the two other probabilities.

Figure 5 shows the evidence on this from our sample of over 150,000 European soccer games (more details on the dataset are below). The left panel shows the percentage of draws and longshot wins with the data having been sorted by deciles of the ex-ante probability of the favourite winning, calculated using the traditional 'overround' method described in Section 2. It shows that as the probability of the favourite winning rises, the probabilities of both draws and longshot wins decline, but the probability of the longshot win falls more than the probability of a draw. The right panel shows that as the probability of the favourite winning increases, the share of the other possible results accounted for by draws rises according to an essentially linear pattern. Intuitively, the more likely the favourite is to win, the more likely it is that the best result the longshot can get is a draw.

In light of this evidence, we calibrate beliefs about the probabilities of draws ($\tilde{p}_D$) and longshot wins ($\tilde{p}_L$) as functions of beliefs about the probability of the favourite winning ($\tilde{p}_F$). Those with subjective belief $\tilde{p}_F$ about the probability that the favourite will win are assumed to have subjective beliefs about the probability of draws and longshot wins given by

$$\tilde{p}_D = (\delta_0 + \delta_1 \tilde{p}_F)(1 - \tilde{p}_F) = \phi_D(\tilde{p}_F) \quad (21)$$

$$\tilde{p}_L = (1 - \delta_0 - \delta_1 \tilde{p}_F)(1 - \tilde{p}_F) = \phi_L(\tilde{p}_F) \quad (22)$$

We use $\delta_0 = 0.338$ and $\delta_1 = 0.405$ to calibrate our model—these are the values from a regression using the data in the right panel of Fig. 5.

4.2 Bettor behaviour

Given the constraint that only unit-sized bets can be placed, optimal behaviour for bettors will consist of taking any bet they consider to have a positive expected value. Technically, the functions $\phi_D(\tilde{p}_F)$ and $\phi_L(\tilde{p}_F)$ are quadratics but, given our calibration, both are decreasing functions over the range of favourite probabilities observed in our model. This means the decision rule for making bets on draws is to make the bet if

$$\tilde{p}_D \geq \frac{1}{O_D} \Rightarrow \phi_D(\tilde{p}_F) \geq \frac{1}{O_D} \Rightarrow \tilde{p}_F < \phi_D^{-1}\left(\frac{1}{O_D}\right) \quad (23)$$

with the corresponding rule for longshots is to make the bet if

$$\tilde{p}_F < \phi_L^{-1}\left(\frac{1}{O_L}\right) \quad (24)$$

These inverse functions, and thus the threshold levels of $\tilde{p}_F$ are uniquely defined over the probability range of $\tilde{p}_F$ being considered. Further details on deriving these decision rules are contained in the [Supplementary Appendix](#).

An important feature of the model is that while the amount of disagreement in this specification for beliefs about the probability of the favourite winning remains constant, the extent of disagreement about the other outcomes depends on the perceived strength of the favourite. Figure 6 shows the level of disagreement about each type of bet: For each value of p_F it reports the gap between the highest and lowest beliefs about the probabilities of the three outcomes. But when the probability of the favourite winning becomes high, disagreement about the probability of a draw becomes larger than disagreement about the probability of a longshot win.

The different levels of disagreement for draws and longshot bets can be explained as follows. Suppose a game has equal probabilities for the three possible outcomes. There will be some people who think the home team has a higher chance of winning and some who think the away team has a higher chance of winning. These bettors will disagree about the

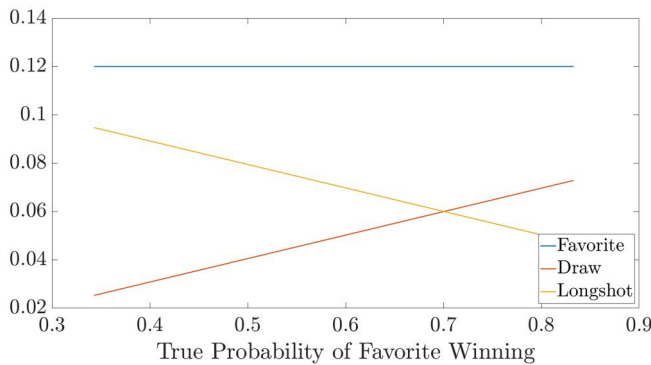


Figure 6. Disagreement (maximum gap) between highest and lowest beliefs for $\sigma = 0.06$.
Source: Authors' calculations.

probabilities of the two teams winning but they will agree on how likely the draw is: Both will think that because their preferred team is a favourite, the draw is somewhat less likely than its true one-third probability. As the probability of the favourite winning increases to a high level, however, there is not much disagreement about whether the longshot can win (almost nobody thinks they can) so disagreement focuses on the probability of a draw and loss rates for those bets rise.

4.3 Optimal odds

In addition to specifying bettor behaviour, because we want to match this theoretical model with the evidence from home/away/draw betting, we also need a realistic calibration of the parameters μ and σ . For μ , which measures non-payout costs per bet for the bookmakers, we can point to some bookmaking markets having margins as low as 3 per cent. As such, we think $\mu = 0.02$ represents a realistic cost rate. For σ , we can use observed values of the overround to calibrate its value. As with the two-outcome version of the model, the average overround in this version moves almost one for one with the value of σ . The average overround for home/away/draw bets in our dataset is 6.9 per cent and this matches well with the model if we assume the value of $\sigma = 0.06$ that we have already used in previous illustrative charts.

With bettors beliefs and other parameters specified, the optimal odds for the favourite follow the same formulas as previously. To solve for optimal odds on longshot wins and draws, we solved the model numerically. Figure 7 shows the predicted odds for favourites, longshots and draws as a function of the probability of each bet's success. Over the full range of probabilities, the most apparent prediction is the obvious one that odds on bets fall as their probability of success rises. However, the apparently tiny differences between the yellow line (odds on longshots) and the orange line (odds on draws) actually generate significant differences in loss rates on these two kinds of bets.

Figure 8 shows the loss rates for the three kinds of bets. Clearly, the odds generally feature a favourite-longshot bias but we also see some interesting patterns for outcomes for betting on draws. When draws are relatively more likely—when the match is effectively considered a toss-up between the three outcomes—the model predicts they will have low loss rates. For these kinds of matches, loss rates for draw bets are predicted to be similar to the loss rates from betting on very strong favourites to win and considerably lower than bets on longshots to win with the same probability of success. But when draws become relatively unlikely, the model predicts that the loss rate from betting on a draw becomes larger than the loss rate on longshots with the same probability of success.

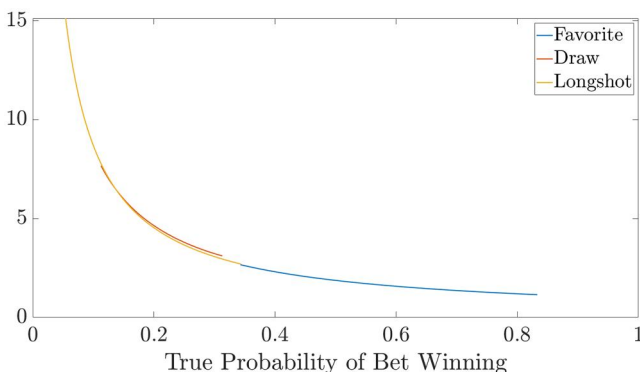


Figure 7. Monopoly odds for $\sigma = 0.06$ and $\mu = 0.02$.
Source: Authors' calculations.

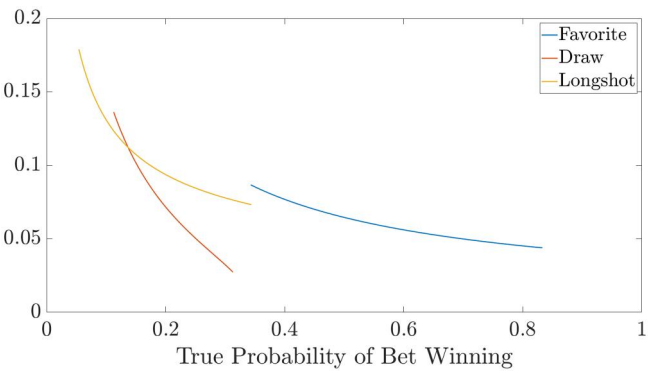


Figure 8. Loss rates for bets on favourite wins, longshot wins and draws for $\sigma = 0.06$ and $\mu = 0.02$.
Source: Authors' calculations.

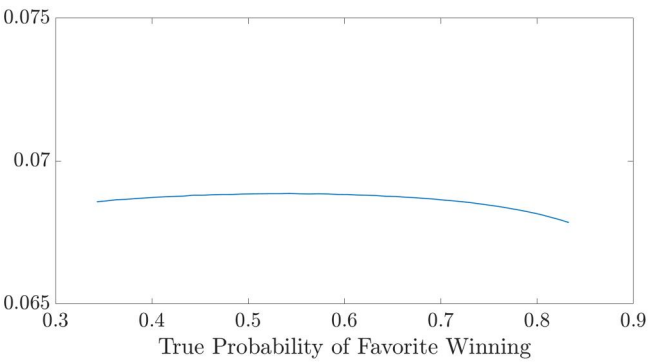


Figure 9. Overround for monopoly odds with $\sigma = 0.06$ and $\mu = 0.02$.
Source: Authors' calculations.

These results are explained by loss rates depending on the level of disagreement as well as the true probability. When there is a very strong favourite, disagreement will be largely about whether there could be a draw, while for games with only weak favourites, the disagreement will be mainly about which team could win. The bookmaker uses the variations in elasticity that come with this varying amount of disagreement to adjust their odds. Another interesting prediction is that while loss rates for bets on longshots should generally be greater than loss rates for favourites, loss rates on teams that are only very marginal longshots are lower than for some of the favourite bets with higher probabilities of winning. Again, this is because the amount of disagreement about the probability of these marginally less fancied teams is lower than for favourites.

Figure 9 shows the overrounds implied by the odds generated by the model. These are relatively flat across all probabilities of a favourite win and, as noted, σ has been set to match the average overround reported in our data. Finally, we don't show the betting volumes predicted by the model but it is worth noting that the model predicts bets on draws will be the least popular option across almost the full range of favourite-win probabilities seen in our data. This appears consistent with anecdotal evidence on sports betting patterns and from data on betting volumes on draws reported on betting exchanges such as BetFair Exchange.

5. Application: European soccer betting

We will apply our model to understanding betting odds on European professional soccer matches. We start with a discussion of the structure of the market for betting on soccer and then introduce our data.

5.1 Market structure in online sports betting

The internet has transformed sport betting and turned it into a huge global business. As betting moved to become predominately online, two different business models emerged.⁶ The traditional retail bookmakers in Europe have adopted what is known as the ‘soft’ bookmaker model. This model focuses on maintaining high gross profit margins and spending on marketing to attract bettors to take on high-margin bets. Well-informed bettors who consistently make profits are generally restricted in how much they can bet and can ultimately be cut off from placing bets.⁷

From a market structure perspective, it is worth emphasizing that there are often significant barriers to entry to the retail bookmaking industry. As discussed in Ed Miller and Matthew Davidow’s excellent 2023 book *Interception: The Secrets of Modern Sports Betting*, retail bookmakers need to make significant investments in either developing their own software for taking bets, making payouts and tracking customers or else buying such software from one of a small number of specialized firms. Bookmakers also face significant regulatory and legal issues relating to handling of clients’ money, such as Know Your Customer rules aimed at avoiding them being used to launder money. Another piece of evidence of market power in this sector, suggested to us by a referee, is that multinational bookmakers expanding into new countries tend to take over existing firms because people seem to prefer to bet with ‘local heroes’.

Taken together, we think these points, while anecdotal, point to retail European bookmakers being a business where operators have market power and where there is little appetite among the firms for competition on price. Importantly for our purposes, these bookmakers dominate the long-established market for betting on home wins, away wins or draws.

In contrast, ‘sharp’ bookmakers have a business model that is essentially the opposite of the soft bookmakers. They set highly competitive odds with low profit margins per bet and focus on making larger profits by attracting high betting volumes. They do not spend much money on advertising and accept bets from informed bettors and professional betting syndicates, using the information from these bets to shape their betting odds. Most of the betting on European soccer taken by these bookmakers is in the form of so-called Asian Handicap bets, which we will describe below.

Given the low margins of sharp bookmakers, why do people still place bets with the soft bookmakers? The explanation is that the sharp bookmakers do not have a significant retail presence in Europe. Taxation on betting profits means they can’t offer the rock-bottom margins that are possible when operating from offshore locations so they are not licensed in most European countries, meaning people can only place bets with them indirectly via brokers. In addition, sharp bookmakers such as Pinnacle also earn fees as ‘market makers’, selling their estimated probabilities to retail European bookmakers, allowing them to profit from retail gambling without having to participate directly in this market. These frictions mean most regular gamblers have either not heard of the sharp bookmakers or consider them too difficult to bet with. In contrast, those who take betting very seriously and/or are betting large amounts will make the effort to be able to bet with the sharp bookmakers.

⁶ An excellent introduction to these two models is David Hill’s 2019 article “Requiem for a Sports Bettor” in *The Ringer*. He provides an informative and entertaining account of how the arrival of European bookmakers was changing the face of US sports betting only one year after the repeal of its federal prohibition.

⁷ The soft bookmaker practices of customer profiling and stake restrictions are documented by Grant et al. (2018) and by Davies (2022a, 2022b).

Table 1. The 22 soccer leagues in the dataset.

Nation	Number of divisions	Division(s)
England	5	Premier League, Championship, League 1 & 2, Conference
Scotland	4	Premier League, Championship, League 1 & 2
Germany	2	Bundesliga 1 & 2
Spain	2	La Liga 1 & 2
Italy	2	Serie A & B
France	2	Ligue 1 & 2
Belgium	1	First Division A
Greece	1	Super League Greece 1
Netherlands	1	Eredivisie
Portugal	1	Primeira Liga
Turkey	1	Super Lig

Source: Authors' calculations based on data from www.football-data.co.uk.

We have presented the extreme cases of perfect competition and monopoly. We believe the low-margin sharp bookmakers that dominate the Asian Handicap market come close to the competitive ideal. In contrast, the market for home/away/draw bets that is dominated by the soft bookmakers comes closer to having a monopolistic structure. We also show in the [Supplementary Appendix](#) that a Cournot oligopoly with a small number of bookmakers produces a very similar outcome to the monopoly model.

5.2 Data description

Our data comes from www.football-data.co.uk, a website maintained by gambling expert and author, Joseph Buchdahl. The dataset has information on game outcomes and odds for bets on a home win, an away win and a draw for 151,683 matches spanning the 2005/06 to 2024/25 seasons for 22 European soccer leagues across 11 different nations as described in [Table 1](#). The betting odds are the average closing odds across the various online bookmakers surveyed by Buchdahl.

6. Empirical results

Here, we present our empirical findings for loss rates from bets on home wins, away wins and draws and then describe our results for Asian Handicap bets on the same sample of matches.

6.1 Loss rates for home/away/draw bets

[Figure 10](#) shows the results for average loss rates for bets on favourites, on longshots and on draws in our datasets, with each type of bet sorted into ten deciles by their normalized probability of success. The betting market exhibits a significant favourite-longshot bias pattern, with bets on favourites losing less than bets on longshots, confirming previous findings of [Buhagiar, Cortis, and Newall \(2018\)](#) and [Angelini and De Angelis \(2019\)](#) using similar but smaller datasets.⁸

The specific pattern of the favourite-longshot bias also fits well with the general pattern predicted by the model as illustrated in [Fig. 8](#). Losses rise in a sharp nonlinear pattern as

⁸ Beyond the favourite-longshot bias inefficiency, several papers have examined whether it is possible to make systematic profits from betting on soccer. Early studies such as [Dixon and Coles \(1997\)](#) and [Goddard and Asimakopoulous \(2004\)](#) concluded this was possible. However, more recent studies by [Caruso and Pernagallo \(2021\)](#) and [Winkelmann et al. \(2024\)](#) suggests there is little evidence for persistent inefficiencies that would allow bettors to systematically make profits.

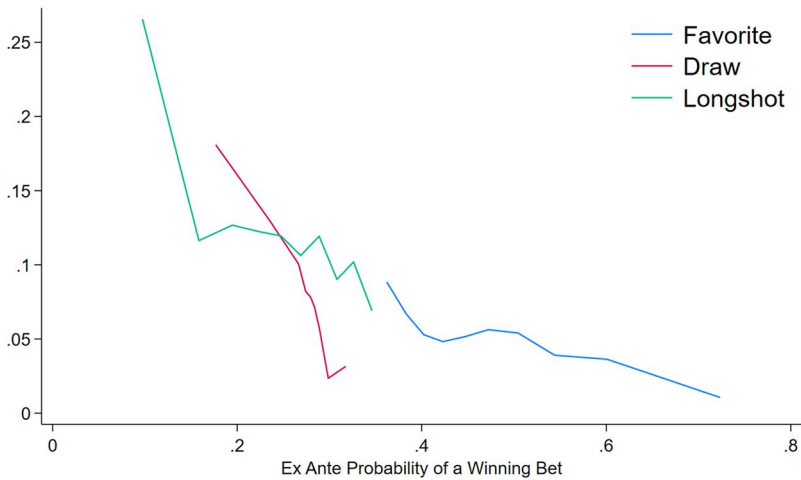


Figure 10. Average loss rates for bets in the home/away/draw market organized by type of bet and ex-ante probability of bet win.

Source: Authors' calculations based on data from www.football-data.co.uk.

the estimated probability of bet success falls. The magnitudes do not fully match our calibrated version of the model—actual losses are a little bit higher than predicted when the probability of success is relatively low and a little bit lower when the probability of success is relatively high—but the size of loss rates generally fits well with the model's predictions.

More importantly, however, the results also conform well with the model's specific predictions about loss rates from betting on draws and longshots. The data conform to the model's prediction that bets on draws with a comparatively high probability of occurring will have some of the lowest loss rates but that bets on draws will have higher loss rates than bets on longshots when draws are relatively unlikely to occur. The data also match the model's predictions that longshot bets with the highest chance of winning have slightly lower loss rates than bets on the weakest favourites.

We should note here that we have set the parameters σ and μ to match the average over-round rather than to match any specific feature of the odds on favourites, longshots or draws—the close relationship between the loss rates in the theoretical Fig. 8 and the empirical Fig. 10 is not the result of any attempt to match how loss rates on different types of bets vary by ex-ante probability of success. However, a simulated method of moments estimator applied to estimating σ and μ based on having the model match the odds for each bet in the sample as best as possible, given the estimated value of the probability of the favourite winning, produced similar estimates of our calibrated values and predicted essentially the same pattern for loss rates.

6.2 Asian handicap bets

Our dataset also contains Asian Handicap odds for the same set of 151,683 matches. As noted above, payouts on these bets depend on an adjustment of the match result that applies a deduction to the goals total of the team considered more likely to win. The Asian Handicap market features bets with handicaps that change in increments of 0.25 goals. Obviously, teams can't score a quarter of a goal, so bets at quarter-goal handicaps are 'hybrids' in which money is split between bets at other handicaps. To see how this kind of betting works, consider first the case in which the Asian handicap is 1.5 and O_S is the decimal odds for the strong team and O_W is the decimal odds on the weak team. In this case, there are only two possible outcomes:

- The stronger team wins by 2 or more. In this case, the bet on the stronger team pays out O_S and the bet on weaker team loses in full.
- The stronger fails to win by 2 or more. In this case, the bet on the weaker team pays out O_W and the bet on stronger team loses in full.

Alternatively, for the case in which the Asian handicap is 1, there are three possible outcomes

- The stronger team wins by 2 or more. In this case, the bet on the stronger team pays out O_S and the bet on weaker team loses in full.
- The stronger team wins by 1. In this case, bets on both teams are refunded.
- The stronger team fails to win. In this case, the bet on the weaker team pays out O_W and the bet on stronger team loses in full.

Bets with a handicap of 1.25 place half the money on a bet with a handicap of 1 and the other half on a bet with a handicap of 1.5, while bets with a handicap of 0.75 put half the money on a bet with a handicap of 1 and the other half on a bet with a handicap of 0.5. Both of these hybrid bets feature the possibility of half the money being refunded, so the only bets that do not feature the possibility of at least a partial refund are those with half-goal handicaps

To compare the behaviour of Asian Handicap odds with those from the home/away/draw market, we need to calculate the probabilities of the different outcomes. For handicaps ending in 0.5, where there are only two possible outcomes, one can use the standard ‘overround’ method described in Section 2. However, when refunds are possible, calculating probabilities is more complex. Consider the case where the Asian handicap is 1. In this case, we want to calculate the probabilities for the three different outcomes

$$P_{S2} = \text{Probability the stronger team wins by 2 or more} \quad (25)$$

$$P_{S1} = \text{Probability the stronger team wins by 1} \quad (26)$$

$$P_W = \text{Probability of a draw or the weaker team winning} \quad (27)$$

Again assuming the expected payoff for all \$1 bets is μ , then we have the following conditions hold.

$$P_{S2}O_S + P_{S1} = \mu \quad (28)$$

$$P_WO_W + P_{S1} = \mu \quad (29)$$

$$P_W + P_{S1} + P_{S2} = 1 \quad (30)$$

This is a system of three linear equations in four unknowns (the three probabilities and the expected return) so there is no unique solution.

Hegarty and Whelan (2025) approach this problem by setting P_{S1} equal to the sample average of the fraction of matches that end in a refund for each kind of handicap. They show that for each of the three types of Asian handicap bets with a refund element, the fraction of bets that end in refunds is stable over time and does not depend on observable match-specific factors such as the betting odds quoted.⁹ Using this approach, for example, for the case with handicap of 1, conditional on a specific value of the probability of a refund, P_{S1} , the other unknowns can be solved to give

⁹ See the Appendix for details.

$$P_{S2} = \frac{(1 - P_{S1})O_W}{O_S + O_W} \quad (31)$$

$$P_W = \frac{(1 - P_{S1})O_S}{O_S + O_W} \quad (32)$$

$$\mu = P_{S1} + \frac{(1 - P_{S1})O_S O_W}{O_S + O_W} \quad (33)$$

The same method can also be used to calculate probabilities of full bet wins for the ‘hybrid’ bets that split odds between integer and half-goal handicaps.

Spread bets offered on high-scoring US sports are generally set to equate the odds of each side of the bet winning. This means the implied probabilities of success of each bet are equal and the odds offered on each bet are typically the same. However, this is not the case with the Asian Handicap market and the implied probabilities of full bet success that we calculate range from about 0.25 to about 0.55.

There are for a few reasons for this widespread in implied probabilities. First, handicaps are only set in quarter-goal increments and these will rarely correspond precisely to the market’s expected goal difference. This means bettors will generally think a bet on one of the teams in a match is more likely to win than the other, which will be reflected in differing odds. Second, for US spread bets, bookmakers normally react to new information by adjusting the handicap while leaving the odds fixed while Asian Handicap bookmakers normally adjust the odds and leave the handicap fixed. Third, the hybrid quarter-point handicap bets have the feature that one side of the bets earns a profit in two of the three possible outcomes, while the other side only makes a profit in one of the three outcomes. For both sides of such bets to be equally attractive, the expected payouts must be the same. This compensation occurs via bets that only make a profit in one outcome tending to have a higher probability of a full payout and this contributes to the opposing sides of these bets having different chances of success.

Figure 11 shows the average payouts for \$1 bets sorted by the estimated probability of the bet winning a full payout. There is no clear pattern of bias across estimated probability ranges and the average returns vary much less than for home/away/draw bets. Because of the handicap adjustment, this market features fewer extreme longshot or extreme favourite bets. This can be seen in the narrower range of estimated probabilities of bet success. The average probabilities of full bet success range from 0.26 in the bottom decile to 0.53 in the top decline, compared with a range of 0.14 to 0.62 for the home/away/draw market.

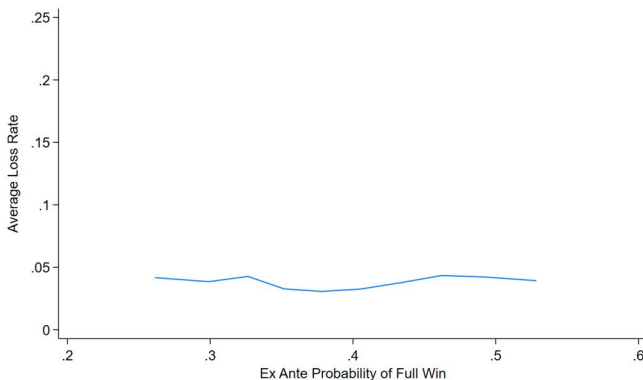


Figure 11. Average loss rates for bets in the Asian handicap market organized by ex-ante probability of a full bet win.

Source: Authors’ calculations based on data from www.football-data.co.uk.

This narrower range of probabilities, however, is not the explanation for the difference in payout rates between the handicap market and the traditional market. The handicap market still features a fairly wide range of ex-ante probabilities of bet success and when comparisons are made over the same probability range, there is a notable difference between the patterns for loss rates in the two markets. For home/away/draw bets, those bets in the decile with an estimated average probability of success of 0.26 have a loss rate of 10.8 per cent while bets in the decile with an estimated average probability of success of 0.47 have a loss rate of 5.1 per cent, so across this range of probabilities, loss rates on longshots are more than twice as high as on favourites. Figure 11 shows that across the same range of probabilities for the Asian Handicap, average loss rates are essentially the same in each decile.

These results thus confirm our prediction that a market in which bookmakers compete with each other on odds will not feature favourite-longshot bias.

7. Discussion

7.1 Deviations from risk-neutrality

We have described a model with risk-neutral bookmakers and bettors but this has largely been for presentational simplicity. In relation to bettors, in the [Supplementary Appendix](#), we discuss the possibility that bettors have relative risk aversion values $\gamma \neq 0$. We use a second-order approximation to the utility function to show that for expected utility maximizers, the criterion for being willing to wager a fraction θ of wealth on a bet with decimal odds of D with subjective probability of success of p is

$$\gamma < \frac{2}{\theta} \frac{pD - 1}{p(D - 1)^2 + (1 - p)} \quad (34)$$

Obviously, if $\gamma = 0$, so people are risk neutral, this collapses to the requirement that $pD > 1$ which we have used up to now. But we can also use this expression to trace out the combinations of decimal odds and subjective probability beliefs that will lead people with different levels of γ to accept bets.

Figure 12 provides examples of demand curves for bets implied by this criteria. It relates to bets worth 1 per cent of wealth ($\theta = 0.01$), in line with the evidence that most people only put relatively small amounts at risk when betting. For relatively small stakes bets of this sort, the second-order approximation to the utility function underlying Equation (34)

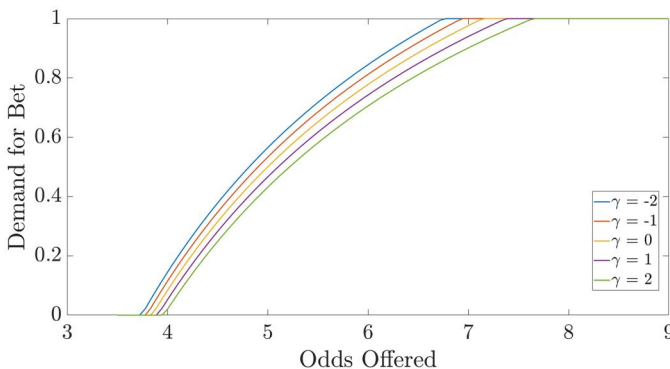


Figure 12. Demand for bets worth 1 per cent of wealth ($\theta = 0.01$) with success probability of $p = 0.2$ at various odds for different levels of relative risk aversion ($\sigma = 0.2$).

Source: Authors' calculations.

is accurate. It shows the fraction of people that will accept a bet of this size with probability of success $p = 0.2$ when beliefs are uniform over $[0.14, 0.26]$, for various different decimal odds. The chart considers relative risk aversion parameters going from $\gamma = -2$ (implying a convex, risk-loving utility function) to $\gamma = 2$ (indicating a strongly concave risk averse utility function). Increasing risk aversion reduces the fraction of people that will bet but the overall shape of the demand functions is very similar. Importantly, our key point that elasticity of demand depends on the probability of a bet's success is robust to having either risk averse or risk-loving bettors.

In relation to bookmakers, incorporating them being risk averse further re-enforces our results. In the [Supplementary Appendix](#), we describe an extended version of the home/away/draw model when the monopolist is risk averse. Profits earned from longshot bets have a higher variance than profits earned on favourite bets, so risk aversion provides another reason for monopolists to set odds with a favourite-longshot bias. For this reason, this version of the model predicts somewhat larger loss rates, particularly for longshot bets. However, once the disagreement parameter, σ , has been set to again match the average value of the overround in the data, the predictions of version of the model with risk aversion are almost identical to the risk-neutral version, though the predicted loss rates for longshot bets and draws are a bit higher and thus match the data somewhat better. For the Asian Handicap market, even if the bookmakers are averse to risk, the high level of competition in this market (including very price-sensitive bettors) rules out the option of them setting particularly bad odds on longshots.

7.2 On beliefs and betting

An obvious question to ask about the gamblers in this model is why they don't use the market price to infer the correct odds. While the monopolistic market differs from the competitive case where you can calculate the probabilities from the odds, there is still a monotonic relationships between the probability of a bet winning and the odds quoted by the bookmaker. Shouldn't those who have a strong belief that one team will win change their mind when the bookmaker's odds show the team is unlikely to?

This question is perhaps best answered by the very existence of betting markets. Bookmakers make profits. Using the market odds as a guide to estimate the true probabilities will generally mean accepting that all bets have a negative expected return. And most people do not bet, most likely for that reason. But some people do bet. This may partly be fuelled by the utility they get from placing a bet, perhaps in a way that increases their enjoyment of watching sporting events. But even then, there is the question of why they choose to bet on one team rather than the other. When one person bets on the home team and another person bets on the away team, the explanation for their differing decisions is likely that they disagree with each other about the probability of their picks winning.

8. Conclusions

Given the increasing size and importance of the online sports betting industry, it is important to understand how it sets prices. We have provided theoretical and empirical evidence that the formation of betting odds varies depending on the structure of the betting market. We have also illustrated how, in less competitive bookmaking markets, the loss rates on bets depend on the elasticity of demand and that differences in elasticities of demand across bets can be explained by a relatively simple model of disagreement among bettors.

Our model shows how favourite-longshot bias can emerge from odds set in a betting market where bookmakers have some market power, without assuming some of the popular 'demand-side' mechanisms that have often been invoked to explain these biases, such as bettors enjoying high-variance risky gambles or having a systematic inability to judge probabilities correctly. These explanations may still play some role in explaining the favourite-

longshot bias but, if so, market structure must also play an important role. For example, if risk-loving preferences among bettors allowed some bookmakers to make especially high profits by offering poor odds on longshots, this raises the question of why the high profits on these bets are not competed away by new entrants. Conversely, the focus on bettors' preferences does not explain why we did not find any favourite-longshot bias in Asian Handicap betting on soccer (although it may be that the participants in that market have different preferences to the typical home/away/draw bettor.)

Our findings are relevant for the emerging US sports betting market. The US market has attracted heavy involvement from European bookmakers, either via direct entry such as Paddy Power owner Flutter's acquisition of FanDuel or via selling services to newly-licensed US bookmakers to allow them to copy the European business model. The rush to enter the market after the 2018 Supreme Court decision and the expensive state-by-state licensing process has combined to produce a market that is dominated by two firms. These firms (FanDuel and DraftKings) began in the pre-legalization era as fantasy sports sites and thus had access to lists of fans likely to bet on outcomes of sporting events.

Media reports suggest the two leading firms have over 70 per cent of the current market, so price setting seems unlikely to meet the ideal of a perfectly competitive market and may be closer to the monopolistic market discussed in our article. These firms also promote long-odds 'moneyline' bets as well as 'parlay/accumulator' and other multi-leg proposition bets that have very low probabilities of succeeding. Our model suggests this market is likely to become characterized by high profit rates and large loss rates for people taking these bets.

Supplementary material

[Supplementary material](#) is available at the *Oxford Economic Papers* journal online. These are the data and replication files and the [online appendices](#) and the data replication files for the theoretical and empirical results.

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